Sample Mark Scheme: P000293

NCFE Functional Skills Qualification in Mathematics at Level 2 (501/2324/5)

Activity 1		Marks
1A1	**CAO (correct answer only) £20 (units required)**	**1**
1A2	**CAO £16.15 (units required)**	**1**
1A3	**£21.25 (units required)**	**2**
	17/0.8 or equivalent	1
	CAO £21.25 (units required)	1
1B1	**197 with a valid check**	**3**
	1576/8 OR their total/8	1
	CAO 197	1
	Check using reverse calculation, for example, 197*8 = 1576 **FT (follow through)**	1
1B2	**(£) 1497.20 (2 decimal places required)**	**2**

Sample

Activity 1		Marks
	1576*0.95 OR 1576*95/100 (FT their value from 1B1)	1
	(£) 1497.20 (2 decimal places required) FT	1
1B3	**116**	**2**
	CAO 116	1
	Check using reverse calculation, for example, 116 +151 = 267 FT	1
1C1	**20 (%)**	**3**
	360 - (90 + 120 + 78) OR 72 seen	1
	72/360*100 OR 0.2*100 OR their/360*100 or equivalent	1
	CAO 20 (%)	1
1C2	**1/3**	**2**
	120/360	1
	CAO 1/3	1
1C3	**(£) 1458**	**3**
	9720/360 OR 27 seen OR 54/360 (=0.15) OR 0.15 OR 15% seen	1
	9720/360*54 OR 27*54 OR 9720*0.15 or equivalent	1
	CAO (£) 1458	1

Sample

Activity 1		Marks
	Total marks:	**19**

Activity 2		Marks
2A	**32 small and 91 large**	**4**
	200/12 OR 16 (small pots)	1
	CAO 32 (small)	1
	13*7 seen OR (140-24)/15 (= 7) multiplied by 200/15 (= 13) or equivalent	1
	CAO 91 (large)	1
2B1	**7.9 m OR 790 cm OR 7900 mm (units required)**	**3**
	1.9 (m) OR 1.3 (m) seen OR 200 - 10 OR 140 - 10 or equivalent	1
	1.4 + 1.4 + 1.3 + 1.9 + 1.9 seen or equivalent	1
	CAO 7.9 m OR 790 cm OR 7900 mm (units required)	1
2B2	**240500 (cm^3)**	**3**
	185 cm OR 92.5 cm used as length	1
	Method 185*130*10 OR 2 (92.5*130*10) FT their sand rectangle lengths but must be internal volume	1
	CAO 240500 (cm^3)	1
	Total marks:	**10**

Sample

Activity 3		Marks
3A1	**28.26 (m^2)**	**2**
	3.14*3*3 OR 3.14*9	1
	CAO 28.26 (m^2)	1
3A2	**18.84 (m)**	**2**
	3.14*6 OR 2*3.14*3	1
	CAO 18.84 (m)	1
3B1	**300 (ft^2)**	**3**
	Area calculation seen, for example, 30*12 OR 10*6 OR 12*10	1
	360 - 60 OR (30*12) - (10*6) or equivalent, for example, the sum of 3 correct rectangles	1
	CAO 300 (ft^2)	1
3B2	**7.28 (kg) with 4 x 2 kg packets selected (with one other amount seen)**	**4**
	208*35 OR 7280 (g)	1
	7280/1000 OR 7.28 (kg) FT	1
	Second correct and sufficient combination, for example, (£) 54.50 for 5 (kg) and 2 x 2 (kg) OR (£) 57 for 2 x 5 (kg) FT	1
	CAO 4 x 2 kg packets (cost £52) FT	1

Sample

Activity 3	Marks
Total marks:	**11**

Overall marks:	**40**
Pass mark:	**26**

Summary of Skills Standards and Coverage and Range

(Note: where task reference and marks are indicated against a skill standard they can be for any of the associated coverage and range statements)

Skills standards	Total marks	Required weighting	Actual weighting	Coverage and range (can be covered across all skills standards)	Task reference	Marks awarded
Representing **R1** understand routine and non-routine problems in familiar and unfamiliar contexts and situations **R2** identify the situation or problems and identify the mathematical methods needed to solve them **R3** choose from a range of mathematics to find solutions	**13**	**30-40 %**	**32.5%**	**a.** understand and use positive and negative numbers of any size in practical contexts	**1A3, 1B2, 2A, 3B2,**	**4**
				b. carry out calculations with numbers of any size in practical contexts, to a given number of decimal places		
				c. understand, use and calculate ratio and proportion, including problems involving scale	**1A1, 1A2, 1A3, 1C1, 1C2, , 1C3, 3A1, 3A1, 3A2, 3A2, 3B2**	**11**
				d. understand and use equivalencies between fractions, decimals and percentages		
				e. understand and use simple formulae and equations involving one or two step operations		
Analysing **A1** apply a range of mathematics to find solutions **A2** use appropriate checking procedures and evaluate their effectiveness at each stage	**13**	**30-40%**	**32.5%**	**f.** recognise and use 2D representations of 3D objects	**2A, 2A, 2A, 2B1, 2B1, 2B1, 2B2, 2B2, 2B2, 3B1, 3B1, 3B1, 3B2,**	**13**
				g. find area, perimeter and volume of common shapes		
				h. use, convert and calculate using metric and, where appropriate, imperial measures		
				i. collect and represent discrete and continuous data, using ICT where appropriate	**1C1, 1C1, 1C2, 1C3, 1C3, 3B2, 1B2**	**7**
				j. use and interpret statistical measures, tables and diagrams, for discrete and continuous data, using ICT where appropriate		

Sample

Interpreting **I1** interpret and communicate solutions to multistage practical problems in familiar and unfamiliar contexts and situations **I2** draw conclusions and provide mathematical justifications	**14**	**30-40%**	**35%**	**k.** use statistical methods to investigate situations **l.** use probability to assess the likelihood of an outcome	**1B1, 1B1,1B1, 1B3, 1B3**	**5**
Total marks:	**40**					**40**

Sample